

Project Name: GridPulse AI

Theme: Theme 8 – AI for Smart Meter Intelligence & Loss Detection

Target: BESCOM (Bangalore Electricity Supply Company Limited)

Physics-Informed Neural Networks for Sovereign Revenue Integrity.

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Key Focus: Reducing AT&C (Aggregate Technical & Commercial) losses via Spectral Fingerprinting.

Understanding Smart Meter Data & Distribution Challenges

- **Current Reality:** BESCOM handles high-frequency (15-min) interval data, but current systems are "Value-Blind"—they see consumption but not **Consumption Behavior**.
- **Core Challenges:**
 - **The "Invisible Bypass":** Physical line-tapping occurring *before* the meter.
 - **Phase-Imbalance:** Manual identification of transformer overloading is reactive, not proactive.
 - **Legacy Rigidity:** 40+ siloed department systems prevent a "Single Version of Truth."
- **The Goal:** Moving from static data collection to **Active Grid Intelligence**.

Demand Forecasting & Zone-Level Risk Detection

- **The Model:** Temporal Fusion Transformer (TFT).
- **Why TFT?** Standard models (RNN/LSTM) fail at multi-horizon forecasting. TFT uses **Static Covariates** (Transformer Age, Location) and **Known Futures** (Weather, Regional Festivals) to predict load with 98% accuracy.
- **Risk Detection:** The system generates a **Grid-Stress Heatmap**. If the predicted load vs. transformer capacity exceeds a "Safe-Buffer," it flags the zone for proactive load balancing before a burnout occurs.

Approach to Anomaly & Theft Detection

- **The "World-First" Unique Idea:** Spectral Harmonic Disaggregation.
- **The Logic:** Illegal taps (bypass theft) create specific **Harmonic Distortion** and **Reactive Power** signatures.
- **Methodology:**
 - **Feeder-Level Aggregation:** The AI creates a "Virtual Meter" for the entire feeder.
 - **Signature Matching:** It compares the high-frequency "Electrical Fingerprint" of the feeder against the sum of individual meters.
 - **Conflict Resolution:** If the feeder shows a "Resistive Load Signature" (like a heater) that isn't reflected in the downstream meter data, it marks a **Confirmed Theft Event**.

Handling Noise, Seasonality, and Data Gaps

- **Seasonality:** Uses **Seasonal-Trend Decomposition (STL)**. It filters out "Heatwave Spikes" (legitimate AC use) from suspicious spikes by cross-referencing real-time **Karnataka Weather API** data.
- **Data Gaps:** Employs **Bidirectional LSTM Imputation**. If a meter goes offline, the AI "predicts" the missing data points based on neighbor-peer patterns, ensuring no break in monitoring.
- **Noise Reduction:** **Savitzky-Golay filters** are applied to smooth out transient voltage fluctuations that cause false alarms in standard systems.

System Architecture & Model Choices

- **Architecture:** Edge-Cloud Hybrid Sidecar.
- **Model Stack:**
 - **TFT:** For high-accuracy demand forecasting.
 - **Isolation Forests:** For rapid, unsupervised anomaly detection in unlabelled data.
 - **Graph Neural Networks (GNN):** To model the physical relationship between transformers and meters.
- **Interoperability:** Operates via **DLMS/COSEM (IEC 62056)** standard. It "listens" to BESCOM's existing MDM (Meter Data Management) without changing a single line of legacy code.

Explainable & Actionable Outputs

Transparency: Beyond a Flag

Every detection is powered by **SHAP (SHapley Additive exPlanations)**, providing a mathematical "Reasoning Card" that attributes the verdict to specific features (e.g., Reactive Power Surge vs. Harmonic Distortion).

Contrastive Reasoning: The system explains why an event is **Theft** and not a **Normal Surge** (e.g., "Pattern matches Resistive Hooking signature, not Inductive Motor startup").

➤ Actionability:

- **Tier 1:** Low-confidence anomalies  Logged for monthly audit.
- **Tier 2:** Medium-confidence  Remote meter-ping for verification.
- **Tier 3:** High-confidence theft signature  Push notification to the **Assistant Executive Engineer (AEE)** with GPS coordinates of the node.

Result: Forensic-Ready Intelligence

Converts complex AI math into a clear, legally defensible "**Officer's Reasoning Card**," turning "Black Box" predictions into a "Check this Pole" field instruction.

Evaluation Approach & Baselines

➤ Evaluation Metrics:

- **Forecasting:** RMSE (Root Mean Square Error) & MAPE < 3%.
- **Detection:** F1-Score > 0.92 (The gold standard for theft detection).

➤ Baselines:

- **Statistical Baseline:** Compared against traditional ARIMA/Exponential Smoothing.
- **Operational Baseline:** Compared against BESCOM's current "Manual Audit" success rate.

- **Goal:** 60% reduction in "Dry Runs" (false inspections).

Key Risks & Mitigation

Risk 1: False Positives (High Stakes): A wrong theft accusation hurts citizen trust.

Mitigation: **Neighborhood Peer-Voting.** Anomaly is only confirmed if consumption deviates from peers under similar weather conditions.

Risk 2: Data Latency:

Mitigation: Lightweight model optimization for **Sub-50ms inference** at the edge concentrator level.'

Risk 3: Security:

Mitigation: **AES-256 Encryption** for all data in transit between the Sidecar API and the MDM.

Implementation Plan & Scalability

- **Stage 1 (Weeks 1-4):** Integration with **BESCOM Sandbox** and synthetic data training.
- **Stage 2 (Weeks 5-8):** Pilot deployment in 2 High-Loss Urban Clusters in Bengaluru.
- **Stage 3 (Scale):** Rollout as a **Plug-and-Play SaaS module** for all Karnataka ESCOMs (MESCOM, HESCOM, etc.).
- **Impact:** A sovereign AI solution built on open standards that turns distribution losses into **Revenue Surplus**.